Virtualization

Virtualization

- A technology that provides an abstraction of the resources used by some software which runs in a simulated environment called a virtual machine (VM)
- Benefits include better efficiency in the use of the physical system resources
- Provides support for multiple distinct operating systems and associated applications on one physical system
- Raises additional security concerns

Hypervisor

- Software that sits between the hardware and the VMs
- Acts as a resource broker
- It allows multiple VMs to safely coexist on a single physical server host and share that host's resources
- Virtualizing software provides abstraction of all physical resources and thus enables multiple computing stacks, called virtual machines, to be run on a single physical host
- Each VM includes an OS, called the guest OS
  - This OS may be the same as the host OS, if present, or a different one

Hypervisor Functions

- The principal functions performed by a hypervisor are:
  - Execution management of VMs
  - Devices emulation and access control
  - Execution of privileged operations by hypervisor for guest VMs
  - Management of VMs (also called VM lifecycle management)
  - Administration of hypervisor platform and hypervisor software